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Fundamental requirements of Digital Twins for production system in Oil and Gas Industry: A systematic literature review

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ABSTRACT

Context: The oil and gas industry is adopting Digital Twins as a significant step in a continuous digital transformation. A Digital Twin can provide intelligent support to main activities related directly or indirectly to oil and gas production, like operations monitoring, process optimization, failure prediction, simulation of what-if scenarios, and safety improvement.
Situation: Specifications of requirements of a Digital Twin (DT) in the oil and gas domain found in the literature are usually presented informally, utilizing natural and often ambiguous language. Most of the requirements need to be extracted from descriptions of DT characteristics and functionality presented in articles.
Objective: This systematic literature review aims to summarize the existing evidence concerning the requirements of Digital Twins tailored explicitly for oil and gas production systems. By thoroughly analyzing published literature, the study seeks to uncover the requirements, properties, and constraints essential for the successful implementation of Digital Twins in this domain.
Method: Through a systematic literature review, the study focused on rigorously identifying common functionality, ubiquitous characteristics, and some emerging trends related to Digital Twin requirements in oil and gas production systems.
Results: From the initial 939 articles, the review selected 94 relevant studies, focusing on described requirements and on application-specific features of Digital Twins. Among the selected papers, 28 were analyzed and reviewed, focusing on specific requirements for Digital Twin for production systems within the industry, shedding light on 17 functional and 7 non-functional requirements common to many DT specifications and implementations.
Conclusion: Our findings underscore the importance of comprehensively understanding and outlining the essential requirements for Digital Twins within the intricate landscape of production systems in the industry. By elucidating key features and properties of DT, this study contributes significantly to enhancing the efficacy and implementation of new Digital Twins, or the evaluation of existing Digital Twins.

As a result, we have identified some important requirements, specifically in the O&G domain. We analyzed some issues related to the software needs of DTs in the O&G domain, highlighting which are the requirements of a DT usually specified or informally described. This study allows us to identify primary studies in both DT for O&G and Requirements Engineering (RE) fields. Even though the requirements described here have been collected from DT works in the O&G domain, many of these requirements are also applicable to other domains, like many areas of engineering and manufacturing. Finally, it aims to offer a clear understanding of state-of-art industrial DT requirements appliable for many domains, but with more focus and detail dedicated to the O&G domain. Moreover, we aim to identify some gaps and challenges that need to be considered for adopting DTs by the O&G industry.

Contents

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1. Introduction

Digital Twin (DT) is a new paradigm where the potential of the real-time availability of data is opening up new avenues for monitoring and improving operations throughout the life cycle of an asset (either a product or a process). Worldwide, many industries, such as the automobile, aeronautics and Oil & Gas (O&G), are adopting DTs as a digital strategy to manage their assets or optimize their processes on all levels.

First introduced by Michael Greaves in 2002, a DT was defined as such: "A set of virtual information constructs that fully describes a potential or actual physical manufactured product from the micro atomic level to the macro geometrical level" [1].

Another definition is given by Moyne et al. [2], where a DT can be a dynamic digital replica of a physical asset, process, system, or product designed with a specific purpose in mind or a virtual representation of a physical product that encompasses both the physical and digital realms, facilitating the exchange of data and information between the two spaces.

Indeed, the benefits of digital twins can include increased transparency, accelerated product development, optimized maintenance, improved quality and efficiency, predictive maintenance, and preservation of asset value throughout the lifecycle [3] and as such, much effort, technologies, and ideas developed in recent years by the 4th industrial revolution are oriented towards such a goal.

Nevertheless, the adoption of digital twins across industries remains in its early stages, with most existing efforts either theoretical or representing partial implementations. These implementations frequently adhere to varying definitions of digital twins, contributing to the fragmented landscape of DT applications.

In practice, a DT can typically consist of a physical entity, a virtual counterpart, and the connections in between. Since then, this definition has formed a basis for more in-depth studies for various purposes employed across industry, academia, and for diverse DT requirements. Indeed, although we can find distinct DTs, each with specific purposes and characteristics, we can also identify common and shared components, behaviors, properties, and features.

O&G production is one of the most capital-intensive industries; thus, it requires expensive equipment and highly skilled laborers. One of the significant challenges in the upstream Oil and Gas Industry (O&GI) is the implementation of new digital technologies that increase production for less unit cost [4]. Digitalization is emerging as a driver for sweeping transformation across the world. Today, O&GI is facing a greater demand for sustainable climate change accountability, together with a transition from the digital frontier to main activities related to production problems, as discussed in Section 2.1.

Indeed, understanding requirements for DTs in O&GI is necessary because:

- The plant installation is too complex, which makes the understanding and tracking of the DT requirements not simple;
- The complexity and extension of the plant prevent a single supplier provides the whole system solution;
- The combination of several provider solutions comes with the problem of understanding what requirements are being attended to by each piece of software or asset component;
- The multiple providers come with inherent interoperability problems that inhibit the required data flow through the whole installed plant;
- There is a permanent worry in O&G industry about the data silo trap, in which crucial production data become inaccessible in private format repositories. Then, there is a wish for agnostic applications that interchange data in open formats.

However, although there is still no consensual definition of a set of requirements for a generic DT, many works describe a set of characteristics adopted in a specific project or context. Many of these DT requirements are common across projects and even field domains, which justifies a need for further studies to consolidate existing knowledge. There are many studies in recent years aiming to provide answers to some issues concerning DTs, mainly focusing on the characterization of Digital Twins [5], DT architecture [6], data management [7], stakeholders collaborations in DT ecosystems [8], Digital-Twin-based testing [9], commonalities and differences of DTs in different contexts [10], safety [11], Internet of Thing [12], DT modeling [13], interactions of DT with other entities [14] and also many applications of DT technology, like, for example, predictive maintenance using DTs [15,16], and usage of DTs to cyber–physical system testing [9]. In contrast, very little attention has been paid to Requirements issues related to Digital Twins, with some notable exceptions such as the Gemini Principles [17]. Indeed, the study of the requirements of DT can generate insights for the adoption of multiple different models and tools created by vendors. Moreover, there can be no doubt that DT requirements study, definition, and comprehension will play a pivotal role in introducing new technologies into DT.

With this in mind, this paper presents a Systematic Literature Review (SLR) to summarize the existing evidence concerning the Requirements of Digital Twins for a specific domain — the Production System in the Upstream O&GI. We intend to identify and analyze the state-of-art of this topic from descriptions of requirements present in published works. Since such descriptions are presented informally, we need to extract valuable and relevant information from such descriptions by listing the requirements of Digital Twin using natural language and defining a consistent set of requirements related to DTs.

Our work differs from the existing surveys and systematic reviews in the area of Digital Twins. [18] surveys the state-of-the-art of DTs in the industry with a focus on the frameworks and applications, [12] explores enabling technologies and challenges, while [19] compiles DT characteristics and design implications. The main focus of the systematic reviews reported by Jones et al. [5], Semeraro et al. [20] are the concept of DTs and identifying key components. Correia et al. [7] presents a Systematic Literature Review on Data Management issues and also proposed solutions in the DT context. Knebel et al. [21] inspects the current state of cloud and edge adoption by digital twins in the O&GI and evaluates cloud-based solutions for digital twins in this context.

This article's main contribution is identifying the fundamental requirements of a DT in O&GI described in the existing literature. A SLR addressing DT requirements can help to define which DT requirements can be considered ubiquitous for DTs mainly in O&G domain, which properties can be regarded as essential for DTs and which DT requirements could be considered optional or mandatory. Indeed, even those who are not experts can benefit from our results.

The paper is structured as follows. After this introduction, Section 2 presents a general overview of production systems in Upstream, Section 3 brings up the fundamentals of requirements engineering, Section 4 explains the methodology behind the SLR, Section 5 presents the results obtained from the application of the previously described method, shows analysis and discussion of the results, while Section 6 presents some threats to validity of our study, the main contributions of our work and the opportunities for future research.

2. A general view of the production system in the upstream O&G domain

The O&G sector is usually segmented into three basic operational sectors: upstream, midstream, and downstream. Upstream activities involve exploring and extracting oil and gas from reservoir fluids. Midstream operations typically encompass transportation and storage phases. Downstream activities involve the refining and distribution of oil products. In this work, we are concerned with the upstream sector, as shown in Fig. 1, specifically with Exploration and Production (E&P) activities.

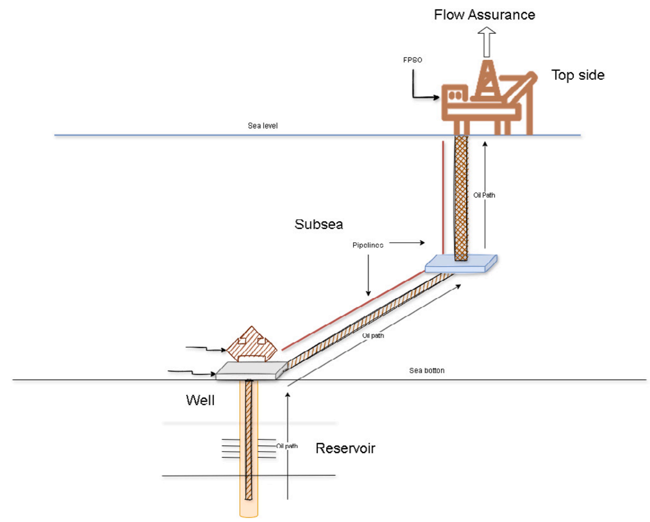

Fig. 1. Upstream sector in Oil and Gas domain.

E&P are the initial phases of the processes chain related to the O&GI. Oil or gas production is a sequence of events starting with the drilling of a well into an underground rock formation, allowing the flow of oil or gas through perforations in the wellbore. Then, the oil and gas travels up the well, passing through a series of choke points and valves. It passes through surface pipelines and treatment facilities until it reaches its final destination in a storage tank or sales terminal [4]. In a wider sense, a production system refers to the mechanism responsible for retrieving subsurface reservoir oil (or gas), transporting it to the surface, processing and treating it, and preparing it for storage and subsequent transportation for further processing and distribution.

Ideally, all companies wish to maximize oil (or gas) recovery from onshore and offshore reservoirs, but actually, their production systems are, on average, running below their maximum production potential due to operational complexities of production and processing facilities. The control room of an offshore platform works at the center of a massive data hub. Data from equipment related to downhole, sub-sea, flow assurance, and topside areas are fed into this hub by many sensors continuously. These sensors control many operational variables – like pressures, temperatures, flow rates, power, vibration, etc. – each with distinct settings, representing many possible control configurations influencing production. Of course, the control tasks are supported by many tools, extensive training, and the expertise of onshore professionals. Control systems and practices are usually adjusted to the capabilities of an asset or a process. However, they do not update dynamically in real time when modifications are made to the asset. The control room operators can also use machine learning to identify bottlenecks constraining production on the platform, like unplanned pressure spikes (slugging), oil-in-water incidents, and unplanned downtime caused by failures in critical equipment.

The O&G companies implement Health, Safety, and Environment (HSE) policies across all operational levels and within every sector. While Health, Safety, and Environment are distinct concerns, each with its own set of technologies, they are frequently integrated into the same functional groups within these oil companies, with adherence to HSE guidelines being a global mandate for operators worldwide. Compliance with the [22,23] standards is usually necessary, the former entails a company's management of its environmental/sustainability goals/policies and the latter adheres to a company's workforce health and safety policies, both also adhering to other external regulations and laws.

Additionally, the internal policies of most corporations establish guiding principles governing operations and the management of risks throughout the entire industry life-cycle.

Historically, intervening in a process in O&GI usually demands extensive manual labor, involving the analysis of multiple engineering documents, spreadsheets, operational manuals, and disparate data sources. This endeavor is time-consuming and error-prone due to the intricate analysis required and the scattered nature of the data.

For example, consider operations on an FPSO: closing a well choke valve, thereby reducing process flow, adjusting injection pump speed that affects pressure in the production system, or engaging the gas lift to boost production. Each requires careful evaluation of adjacent system components, upstream pressure dynamics, and reservoir performance.

Indeed, the O&GI is very data-rich, [24] with different sources (Well treatment data, Production rates, Well testing, Fluid analysis, etc.), formats (time series, spreadsheets, 2D or 3D images, etc.), size, generation rate (Up to several numbers per time unit), and application areas (reservoir engineering, production management, etc.). Such cumulative data evolution is an important characteristic of O&G facilities. Huge quantities of asset data are generated throughout all project lifecycle phases. The great problem is that such data is spread over many software applications, databases, and (digital or physical) files at the enterprise and the site. Consequently, such data could be often inaccurate, out-of-date, incomplete, inconsistent, and/or poorly synchronized.

A DT can provide intelligent support to main activities related directly or indirectly to production, like operations monitoring, process optimization, failure prediction, simulation of what-if scenarios, and safety improvement. DTs will not replace the conventional models and physical understanding of O&G asset operation; they will supplement them, filling in the gaps that hold back production.

2.1. Digital twin in the Oil and Gas domain

As rightly noted by Voas et al. [25], a Digital Twin (DT) can represent either a physical asset or an abstract entity—that is, one without a traditional physical form or manifestation. A fitting example is a process within manufacturing or oil production. In the context of oil and gas (O&G), a digital twin is a complex system comprising components that perform distinct tasks in the production process, such as plant monitoring, maintenance scheduling, asset usage optimization, and human work planning. The system relies on seamless data and information exchange between software components to enable optimized process control. Since this work focuses on oil and gas production systems, the scope of requirements discussed here pertains primarily to Digital Twins that represent processes. To fulfill this purpose, the Digital Twin must realistically and dynamically replicate the behavior of physical assets and processes in real-time. This is typically achieved through dynamic process simulation across the entire range of asset functionality. In essence, the DT functions as a dynamic process simulator.

In an O&G production system, a DT can use advanced technology for sensors, data communication, data analytics, collaboration, and decision-making throughout the value chain, including completions, wells, pumps, pipes, chokes, compressors, and treatment facilities, improving activities like artificial lift, well treatment, fluid separation, equipment maintenance, and HSE processes at lower costs.

The O&GI involves high-risk activities beyond exploration, including the production and processing of flammable and explosive materials. This sector employs intricate and costly machinery, significant manpower, and extensive support systems. Each activity presents different hazards and requires distinct control methods to reduce risks to the lowest practicable level. Hazards include high pressure, temperature, motion (rotating equipment, drill pipes), and electrical risks. Eliminating these hazards is challenging, so HSE processes focus on preventing fires, explosions, and environmental accidents and ensuring personal safety.

Although there is no uniform definition of "intelligent oilfield", an essential aspect of intelligent oilfields is the coordinated development of field production, decision-making, and modern information technology applications [26,27]. Sometimes, other terms are also used to refer to this combination of O&GI elements and the intelligence from digital technologies, like "smart field", "smart well", or "intelligent production system". Several other companies also use these technologies and have been for years, but the adoption of the term "digital twin" is relatively recent. In fact, OnePetro, which is an online library for the E&P industry, has no references to the term "digital twin" before 2016. We find other commonly used terms in other domains representing similar concepts of creating a single version of truth for all data about a physical entity, with some important distinctions, but sharing the same theme of data generation and information management, such as Building Information Modelling (BIM) and Building Lifecycle Management (BLM) in construction industry domain and Product Lifecycle Management (PLM) in manufacturing domain.

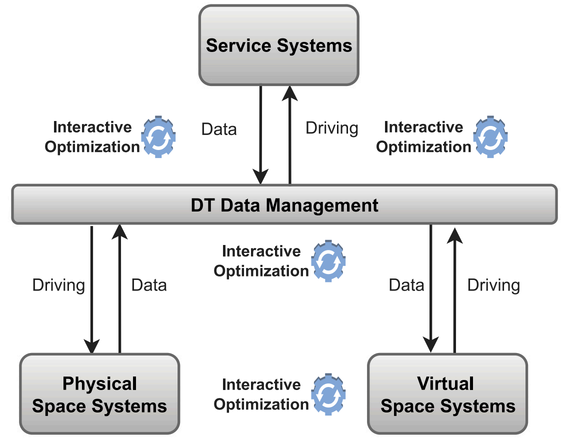

Fig. 2. The 5-dimensional architecture.

2.1.1. Components of digital twin: the 5-dimensional architecture

The literature presents many distinct components and architectures for Digital Twins. The original digital twin concept consists of 3 components: the physical asset in a real space, the digital representation of an asset in a virtual space, and the connections between both, typically through a bidirectional flow of data and information [1].

The emergence of technologies supporting the DT model, such as the Internet of Things, sensors, and data storage and processing capacity, enabled an exponential growth of DTs. As the concept of DT has evolved and industry implementations have matured, many researchers and practitioners have considered requiring other key components in the structure of a DT to meet data capture and analysis needs. The definition of a DT with three dimensions was extended to cover the new requirements and components always present in DTs described in industrial applications. Tao initially described the complete DT model with five dimensions in Tao et al. [28], Tao et al. [18], Qi et al. [29]. According to this model, such dimensions are necessary to support different DT modules: physical entity, virtual model, DT data, services, and connection. In the 5-dimensional architecture, shown in Fig. 2, a data repository integrated into the solution and a service structure to enable data access and analysis methods are essential. This defines the two additional dimensions of the model: DT data and services for unified data management and on-demand utilization.

2.1.2. Digital twin in O&GI: some use cases

DT development is one of the key enablers for successful engineering, integrating, maintaining, and optimizing production systems in the O&GI, which is indicated by the increasing number of related publications in key conferences and journals investigating these challenges, e.g., [30–36]. Moreover, Digital Twins can provide enabling capabilities for optimizing operations, improving safety, and enhancing decision-making within the complex environment of the O&GI.

A DT can provide intelligent tools not only to main **activities related directly to the Production system** but also to **activities supporting the production and providing support for better decision-making**, like, for example:

- To use advanced data analytics techniques to extract meaningful insights from complex datasets to understand reservoir behavior and optimize production;
- To predict equipment failures (predictive maintenance), detecting early signals from failing or degrading equipment, allowing for the planning and implementation of corrective actions before failure occurs and generally at a significantly lower cost. Sometimes, a small failure can lead to a huge failure and a shutdown, requiring a production stop. In fact, predictive maintenance reduces maintenance frequency to the lowest possible state, leading to high resource savings in normal working conditions [37];

- To remotely control and monitor operations, reducing the need for physical presence in off-shore environments;
- To simulate a series of “what-if” future operating scenarios for existing (or new) assets or for some operating conditions and situations;
- To make available advanced visualization tools and dashboards to present an overview of KPIs (Key performance indicators like total wells, gross vs. net well count, equipment downtime, etc.) related to wells;
- To ensure compliance with both industry standards and regulations governing HSE issues and fluid production.

The way these activities are executed can vary based on distinct companies and even specific projects.

Thus, for successfully engineering DT for Production Systems in O&G, fostering research on the Requirements of such DTs is crucial.

3. Requirements for DTs in O&GI: fundamentals

Requirements Engineering (RE) entails the process of gathering stakeholder needs and desires, refining them into a collectively agreed-upon set of detailed requirements, serving as the groundwork for all subsequent development endeavors [38], including several activities like elicitation, representation/modeling, validation, and management [39]. Consequently, RE is one of the most important activities in any product development life cycle since it is the basis for product design and realization.

Like for any system, understanding the requirements of a DT usually provides a good comprehension of its structure and architecture (which are its assets and components, and the relationship among them), its behavior (which are the visible external functionality and the internal functionality supporting it), its goals (either broad and long-term business goals and objectives, specific and measurable users goals and objectives) and its non-functional properties (which describe the constraints on the DT as well as the quality aspects of the DT, and characteristics and factors to be taken in account). These generic properties (also known as Non-functional requirements) define the overall qualities of the DT. Since they restrict system services and behavior, non-functional requirements are often critical, and functional requirements may need to be sacrificed to meet them. In general, the specification of such requirements could be obtained from the DT features currently supported by some actual DTs implementations.

Identifying DT requirements is also difficult due to a lack of standard understanding of DT concepts. Despite some contributions related to DT ontological models [40,41], there is not yet an ontological consensus concerning Digital twin concepts, and indeed, this issue is still an ongoing effort [20].

Thus, the actual adoption of DTs into the O&GI often follows different definitions, having varied requirements and specifications. A DT could be created for a single well to an entire oil or gas field, for an asset or a process, or even as a representation of a facility. It can be used to understand, predict, and optimize performance to help achieve top performance and recover future operational losses by integrating data, simulation, and visualization of the entire operations of the production system, from the subsurface equipment to central processing facilities [42].

Even adopting a well-known DT definition, [5], there is still no consensus on the characterization of DT components and on what are the essential requirements for the operation of DT. Moreover, DT requirements are handled on a proprietary and case-by-case basis without considering generic or reusable requirements. This lack of essential requirements makes it difficult to identify how and which characteristics, processes, and requirements of previous (successful) DTs could be reused in specifying and modeling future DTs requirements. Some conditions are more critical and relevant in a given domain or context. Even within O&G’s upstream environment, different tasks can significantly emphasize other expected functionalities of a DT, such as from a maintenance perspective, where the focus on a 3-D visualization and physical simulation is paramount.

In fact, Digital Twin in O&GI is not a singular emerging technology but rather a convergence of various technologies – such as Big Data, data visualization, analytics, simulation, cloud computing, Internet of Things (IoT), and data interoperability – that allows the manifestation of increased intelligence in O&GI processes. Some of these technologies, such as simulators and data visualization, have been around for a long time. In contrast, others like IoT and Machine Learning are newer and gaining traction with O&GI services.

However, O&GI has some peculiarities [43,44] that should be considered in the implementation of DTs:

- Diversity in Models and Tools: considering the dimensions and complexity of the O&GI, information and models will be generated and managed by many different software tools, and for different purposes, each with its way of structuring and classifying data. A DT application should be able to gather and integrate the largest amount of information that the user can access to the minimum translation or conversion cost that would be considerably high risk (and cost) during the asset lifecycle.
- Agnostic approach: In principle, no technology provider can meet the wide range of requirements for all the information that needs to be part of a digital twin. However, even if a single supplier met all requirements, several experts say relying on a single technology provider is not advisable. Indeed, interoperability has been identified as particularly challenging as, over recent years, DT providers grew fragmented as island solutions. Consequently, the computer environment needs to adopt an agnostic approach in order to provide heterogeneous data integration, aggregating data from diverse sources (reservoir sensors, SCADA systems, IoT devices) spanning structured/unstructured, real-time/historical formats. The key is ensuring (i) a set of standards are defined before adopting some tool, service or data format; and (ii) to establish a quality assurance plan to reinforce these standards for each participant and supplier in creating a DT. More, the digital twin must integrate with legacy systems and older control systems used by existing facilities, which might use different protocols or lack modern IoT capabilities.
- Insurance of data security: As digital twins rely on data connectivity, they become targets for cyberattacks. Ensuring security in data transmission and storage is critical, especially given the critical infrastructure involved. So, DT need to provide restrict access to extremely confidential information since there is a risk of data exposure.
- Extreme Conditions: Oil and gas facilities often operate in extreme conditions – pressures, temperatures, and corrosive environments (e.g., offshore platforms) – and thus DT needs to be designed to work under such conditions.
- Regulatory Compliance: Oil and Gas Industry is heavily regulated, and thus Digital twins might need to incorporate compliance checks, and reporting features to meet environmental and safety standards.
- Scalability: Oil fields with hundreds of wells can be massive, and thus the DT must handle large-scale systems without lagging.
- Remote Operation: Oil and gas facilities are sometimes in remote locations, like offshore or polar regions. Digital twins might be used for remote monitoring and control, consequently reducing the need for onsite personnel. Remote operation would require reliable communication links and, eventually, edge computing for data processing on-site. Moreover, in remote locations, transmitting large volumes of data in real-time might be challenging and DT needs to face data latency and bandwidth issues.
- Cost Optimization: Since production systems need to balance output with costs, the digital twin could simulate different operational scenarios to maximize profitability while minimizing costs.

In the DT context, requirements engineering challenges are complex. Understanding the production process activities and driving forces in any enterprise is essential. This might involve political and cultural aspects as well as socioeconomic factors. On the other hand, the requirements engineer needs to be aware of standards, policies, and regulations on national, and international levels, which are typically expressed by legislation, recommendations, and standards, for example: ISO-30173 [45], and Chen et al. [46].

Another significant challenge is that most of the DT descriptions found in the literature are presented informally, utilizing natural and often ambiguous language and not forming a comprehensive and consistent set of requirements for DTs. Thus, we need to extract useful and relevant information that is considered to be a requirement from informal descriptions of DT characteristics and behavior. To avoid such effort of extraction causing misunderstandings about requirements, some guidelines were adopted in this paper:

- Requirements are written in plain language that preferably reflects the user's perspective and terminology in order to communicate adequately the goals and the needs;
- Requirements are **agnostic**; therefore, all functional requirements are ideally implementation-neutral. In other words, requirements should state **what** the DT should do but not **how** DT should do it.

4. SLR: Planning and execution

As a research area matures, it becomes important to summarize and overview the increasing number of research works. Systematic literature review (SLR) is a method for structuring such scientific landscape [47],employing a protocol to systematically uncover, analyze, and interpret all pertinent evidence concerning a defined research query, ensuring impartiality and reproducibility [48].

Systematic literature reviews are very common in software engineering [49,50], Human-Computer Interaction [51], and even in the domain of O&G [27,52].

In this paper, we present a study done with 939 articles published until October 2022 to understand and describe the requirements in the Digital Twins domain, particularly for Upstream Production Systems in the O&GI. We are also interested in identifying when and where the results are published and by whom.

With DT being a multi-disciplinary, heterogeneous challenge, we consider **requirements** broadly, i.e., we include functional requirements, non-functional requirements, properties, and constraints in our study because they are essential in a requirement specification document as they help to make explicit some stakeholders' expectations and to define some quality properties related to the desired DT.

In summary, the contributions of this paper, hence, are:

- The state-of-the-art of research on Requirements of Digital Twins for Production Systems in the O&GI, resulting of a SLR from the novel and unique primary studies published until October 2022;
- The discussion and investigation of limitations and future work in the Requirement Engineering process for DTs for O&GI in Section 6.

The resulting research landscape can help to understand, guide, and compare research on this subject. In particular, we intend to identify, categorize and structure the main DT requirements addressed by the research community and some challenges that seem less investigated.

In the following, Section 4 details of planning and execution of our research method and Section 5 summarizes the findings.

4.1. SLR planning

We planned this review by proposing a research question relevant to our research objectives and defining the search strategy, search string inclusion/exclusion criteria and quality assessment. A comprehensive elaboration of these elements is provided below. The goal of the study was to examine which requirements were considered during the development of a DT in O&GI.

To achieve the main goal, we set the following research question:

- **RQ1**: What are the requirements of a DT for Production Systems in the O&GI usually specified or (even informally) described during DT development?

Search Strategy: the following four digital libraries were primarily used: ACM, IEEE, ScienceDirect, and OnePetro. While ACM, IEEE, and ScienceDirect are well-known digital libraries in Computer Science and Engineering, OnePetro is an online library of technical literature for the O&GI, containing mainly resources (from journals, conferences, and ebooks) upstream. The search engines used mainly were internal engines of the libraries mentioned above and Google Scholar.

Selection Criteria: The following inclusion (IC) and exclusion (EC) criteria were used:

- IC1: The study must present some example of a DT requirement (an informal description, a rigorous description, a model, or a specification related to DT requirements);
- IC2: The study must present some requirement related to DT for O&G and/or Upstream or Production Systems;
- IC3: The study is relevant to the search terms;
- IC4: The title of the study must contain at least one of the terms related to DTs in the context of the production system;
- EC1: The study is not written in English;
- EC2: The study is about DTs but does not present DT requirements (or DT characteristics or DT properties);
- EC3: The study is not published as a short (2–4 pages) or full paper (more than four pages);
- EC4: The study is a duplicated record retrieved from different search sources;

Quality Assessment (QA): This step should be performed by at least two evaluators who analyze each one of the studies in order to answer the quality assessment questions as follows: yes, partially, and no.

Each of the questions prepared for the quality checklist has a score and a weight according to the question's importance in relation to the research objectives. If a given question is fully answered, it gets the full score; if it is partially answered, it gets half the score; else, its score is zero. The sum of the scores for all these questions was used to assess the quality of the primary articles: only studies that scored 2.0 to 4.0 were considered for further analysis (detailed reading). Since we observed that questions related to DT requirements (QA3) and (O&G) requirements (QA4) held a higher priority compared to questions concerning DT Specification/Modeling (QA1) or representation and modeling of DT requirements (QA2), the quality assessment incorporates weighted values, with QA4 and QA3 being more significant than QA1 and QA2.

Table 1 shows the quality assessment of each question, considering the following weights:

- QA1: Does the article make any contribution to DT modeling or specification?
- QA2: Does the article present any DT requirement representation or model?
- QA3: Does the article describe examples of DT requirements?
- QA4: Does the article describe any characteristic or requirement for DT in O&GI?

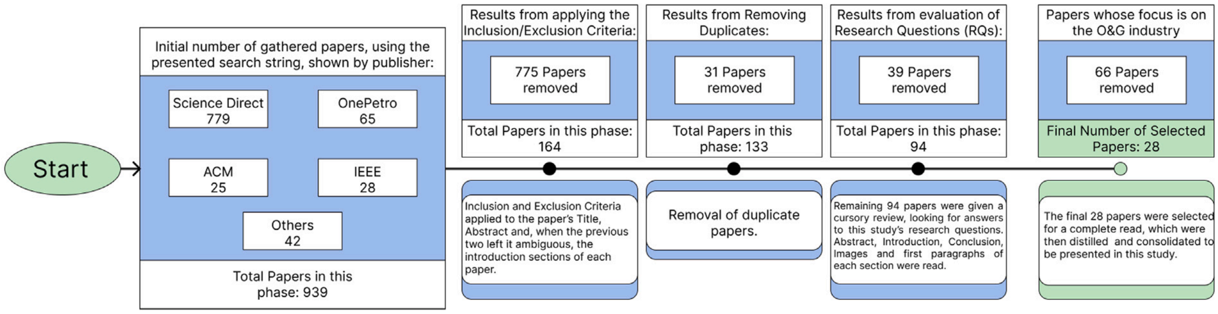

Fig. 3. SLR timeline.

Table 1
Quality Assessment of each question by approval percentage.

Criteria	Grades	Approval Percentage	Score Weight
QA1: Does the article make any contribution to DT modeling or specification?	(No,Partial,Yes)	76,78%	1.0
QA2: Does the article present any requirement representation or model?	(No,Partial,Yes)	82,12%	1.0
QA3: Does the article describe examples of DT requirements?	(No,Partial,Yes)	91,07%	1.5
QA4: Does the article describe any characteristic or requirement for DT in O&GI?	(No,Partial,Yes)	92,85%	2.0

4.2. SLR execution

The search string construction, studies selection, quality assessment, data extraction, and synthesis were performed during this phase. The findings can be accessed in Section 5.

The process of execution of SLR is illustrated in Fig. 3. We sampled papers by identifying relevant works using four different Digital Libraries (DLs) to perform the SLR and snowballed the references from these works and surveys, finding additional relevant studies.

We refined the list of keywords iteratively according to the quality and amount of studies resulting from the DLs search. The search string was composed using three categories of terms: (i) synonyms for the term Digital Twin, (ii) synonyms and similar terms for requirements that can reveal requirements (such as properties, specification, etc.), and (iii) Terms and synonyms related to the O&GI and Upstream.

The search string can be seen below:

((“digital twin” OR “DT” OR “digital twins”
OR “smart field”
OR “digital twin modeling” OR “digital twin specification”
OR “smart well”
OR “production system”
OR “intelligent production system”
OR “digital model”
OR “optimization”
OR “intelligent maintenance”)
AND
(requirements OR characteristics OR properties
OR specification OR modeling OR gathering technique
OR process OR needs OR representation)
AND
(“Oil and Gas Industry” OR “O&G industry”
OR “Oil” OR “Gas”
OR “Petroleum”
OR “Upstream”
OR “Oil Field” OR “Field Production”
OR “Petroil industry”))

The primary studies selection process involved retrieving a significant number of articles from different databases, including ACM(25), IEEE(25), ScienceDirect(778), and OnePetro(65), as well as other diverse publications(41). Through this initial search, we obtained a substantial pool of articles(939) for further analysis.

One of the researchers inspected the studies' titles and keywords (Round 1) by eliminating duplicate pieces and applying the inclusion, exclusion, and quality criteria. As a result of this first classification round, we ended up with 94 candidate studies. Then, in Round 2, the pre-selected studies were assessed by a second (one of the co-authors) researcher into a more in-depth analysis, trying to answer the aforementioned Research Question RQ1, adopting qualitative manual analysis: we studied existing descriptions of characteristics, functions, components, and properties in the O&GI domain in which a DT was proposed to distinguish among different types of requirements. To review the agreements and disagreements raised by the researchers in their assessments, we conducted a face-to-face consensus meeting. For the papers where consensus was not reached, the two researchers read the entire paper to make the inclusion or exclusion decision. This procedure has revealed some aspects that would foreshadow conclusions in the later stages of analysis. For example, 164 articles fit into IC1. However, as seen in the deeper analysis of the RQs, most of them focus on a review of distinct definitions of DTs and a broad discussion of basic DT concepts. Just a few were actually tied to any discussion on the requirements or properties of DTs.

4.3. Overview of studies

Before we present the findings, we present some statistics from the selected studies.

According to Fig. 4, most selected works (939 in total) came from Science Direct (83%), followed by OnePetro (6.8%), IEEE (3%), ACM (2.7%), and 4.5% from other sources. Among the final 28 O&G papers, most (68.97%) were from OnePetro, fitting this study's focus.

Fig. 5 shows the country of selected work origin based on co-authors' affiliation. We identified contributions from 14 countries, with Germany (24.4%) and China (18.9%) as major contributors. Other countries had less than 11% each. For the 28 O&G articles (Fig. 6), Germany led (24.1%), followed by the UK (20.7%) and the USA (17.2%), with a smaller gap between major and minor contributors. German papers (7) focused on a specific DT technology from a well-known supplier, while other countries' works were more agnostic. Most contributors were European, with exceptions like the USA, Canada,

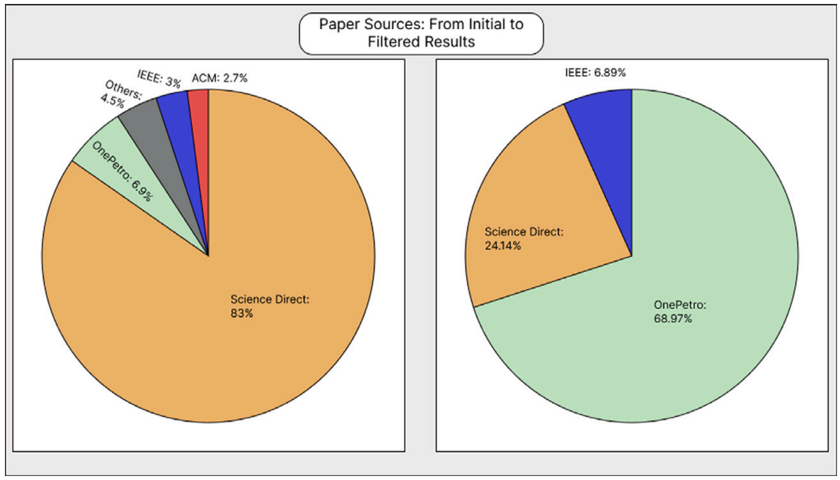

Fig. 4. Paper Sources.

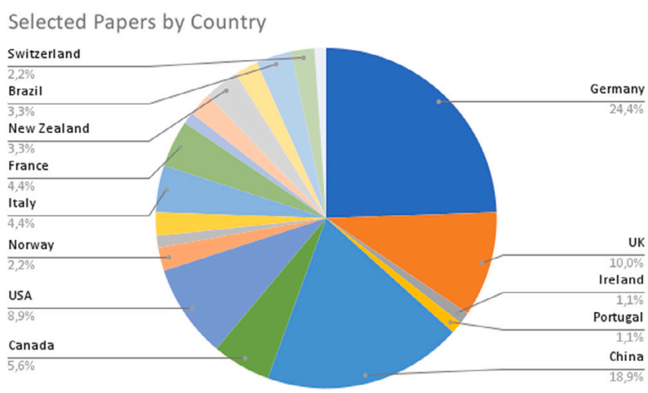

Fig. 5. Provenance of all papers.

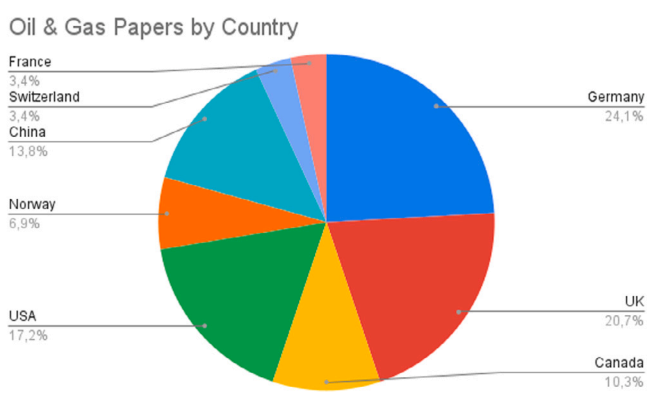

Fig. 6. Oil & gas papers by Country.

Brazil, and New Zealand, which have large companies and academic groups in the DT domain.

However, when looking specifically at the 28 O&G articles — see Fig. 6, the provenance changes; while the majority still belongs to German publications (24,1%), the second contributor is the UK (20,7%), and the third is the USA (17,2%). Moreover, the gap between the major contributors and the other countries is smaller. Investigating the nature of these 28 papers, all publications from Germany (7 papers) described research adopting a specific DT technology from a well-known supplier, while works from other countries were more agnostic. Most of the contributors were European countries, with a few exceptions, such as the USA, Canada, Brazil, and New Zealand. All these countries have big companies in the power, energy and environment industries and also academic research groups working on the DT domain, which may explain these results.

5. SLR analysis and answers to the research question

The surveyed literature revealed some findings about both requirements for DTs and problems and limitations occurring during the Requirements Engineering process. Due to the typical and inherent confidentiality of information concerning the O&GI, we understand most domain-specific requirements sets, and discussing actual problems and limitations is often unavailable.

Out of these 94 selected articles, most (75 papers) studies answered RQ1, usually with an informal description of essential DT characteristics and application-specific DT features and services. For the purpose of this research, we use the term "requirement" when the paper mentions functional needs, external behavior, non-functional requirements, quality criteria, composing elements (components), features, characteristics, properties, and constraints. Such a description provides a broader view of DT technology and generic DT requirements that can be more broadly applied to the O&G domain.

For the study, only 28 O&G papers focused on a more thorough review, presented in Table 3. Concerning the research question, out of these 28 papers, the totality of studies, namely 28 of 28 (100%) papers, answered RQ1.

We surveyed the selected papers to find descriptions of the requirements of DTs, specifically requirements for production systems in the O&GI. In general, such descriptions are presented informally, listing the requirements of DTs using natural language. Since some publications are not requirements-oriented, they may not describe exactly the requirements, and thus, we need to extract useful and relevant information from descriptions of DT characteristics and functionality.

To allow further content analysis and assess central aspects of the studies defining fundamental requirements for DTs, we adopt a classification scheme with five categories for requirements, following a conceptual architecture framework for DTs in the upstream O&G industry (Leipnitz, M.T. et alli. 2024), based on the five-component architecture proposed by (Tao and Zhang, 2017). We think such conceptual architecture captures domain-specific needs and serves as a blueprint for organizing the DT responsibilities. Leipnitz's paper explains such architecture of Digital Twin to make clear the concepts and fundamentals, since it intends to group functional components to support the modular construction of a complex DT, combining

Table 2
Quality criteria score of the selected articles.

Reference	Paper Title	QA1	QA2	QA3	QA4
[33]	Highly Scalable Digitalization Platform for Oil and Gas Operations Enables Total Asset Visibility for Predictive, Condition-Based Fleet Management Across Single and Multiple Sites	No	Yes	Yes	Yes
[32]	How to Achieve Project and Operational Certainty Using a Digital Twin	Partial	Yes	Yes	Yes
[31]	Developing a Digital Twin: The Roadmap for Oil and Gas Optimization	Yes	Yes	Yes	Yes
[36]	Digital twin for the oil and gas industry: Overview, research trends, opportunities, and challenges	Partial	Partial	Partial	Yes
[53]	Improved Decision-Making, Safety and Reliability with Visual Asset Performance Management	Yes	Partial	Yes	Yes
[54]	Oilfield virtual twin	Yes	Partial	Yes	Yes
[55]	Using “Digital Twin” Of Coriolis Meters For Multiphase Flow Measurement	No	No	Partial	Yes
[56]	Starting with the end in mind: The path to realizing digital value	No	Partial	Partial	Yes
[57]	Digital twin of subsea pipelines: conceptual design integrating IoT, machine learning and data analytics	Yes	Partial	Yes	Yes
[58]	The evolution of asset management: Harnessing digitalization and data analytics	Yes	Yes	Yes	Yes
[59]	How digital engineering and cross-industry knowledge transfer is reducing project execution risks in oil and gas	Yes	Partial	Yes	Yes
[60]	Updated case study: The pursuit of an ultra-low manned platform pays dividends in the north sea	No	No	Yes	Yes
[61]	Machine learning and artificial intelligence as a complement to condition monitoring in a predictive maintenance setting	No	No	Yes	Yes
[62]	Improved collaboration and transparency of oil & gas capital projects using a digital project delivery system	Yes	Partial	Yes	Yes
[30]	A roadmap for adopting a digital lifecycle approach to offshore oil and gas production	Yes	Yes	Yes	Yes
[63]	Digital twin bridging intelligence among man, machine and environment	Yes	Yes	No	Partial
[64]	The Dawn of the New Age of the Industrial Internet and How it can Radically Transform the Offshore Oil and Gas Industry	Yes	Yes	Yes	Yes
[65]	LIVE Digital Twin for Smart Maintenance in Structural Systems	Yes	Yes	Partial	No
[66]	Machine learning based digital twin framework for production optimization in petrochemical industry	Yes	Yes	Yes	Yes
[67]	Modeling methodology for application development in petroleum industry	Yes	Yes	Partial	Partial
[68]	The digital twin of oil and gas pipeline system	Yes	Yes	Yes	Yes
[69]	Using an advanced digital twin to improve downhole pressure control	Yes	Partial	Yes	Yes
[70]	Asset management of oil and gas pipeline system Based on Digital Twin	Yes	Yes	Partial	Yes
[34]	Applications of Digital Twins to Offshore Oil/Gas Exploitation: From Visualization to Evaluation	Yes	Partial	Partial	Partial
[71]	Digital Transformation through Integrated Power & Process	Partial	Yes	Yes	Yes
[72]	Digital twin for operations-present applications and future digital thread	Partial	Yes	Partial	Yes
[73]	Digital twin design requirements in downgraded situations management	Partial	No	Yes	Yes
[74]	Exploiting the Full Potential in Automated Drilling Control by Increased Data Exchange and Multi Disciplinary Collaboration	Yes	Yes	Yes	Yes

several providers’ solutions and defining scope and responsibilities. This architecture meets the elicited generic properties of DTs 1 to 3, described in this paper: scalability, modularity, and interoperability. The functional components are the application components (frontend, data storage, backend), and infrastructure components (orchestration, containerization, authorization, authentication, and security).

The five categories correspond to components in the five-component architecture proposed by Tao and Zhang [75], Qi et al. [29], Tao et al. [18], Tao et al. [28] — physical entity, virtual model, data, connection, and services, described in Section 2.1.1. We organize the selected papers into 5 categories: each category corresponds to one component of this architecture – as shown in 3 – where for each paper, we mark with character **V** if the paper discusses some requirement related to the respective dimension: physical entity, virtual model, DT data, services and connection, and we mark with character **X** otherwise.

Thus, requirements are classified as

- requirements related to the physical entity;
- requirements related to virtual model;
- requirements related to data;
- requirements related to connection;
- requirements related to services.

The selected twenty-eight (28) articles were analyzed and fully reviewed, focusing on the intersection between Requirements, Digital Twins, and the O&G domain.

The high variability of the components involved in a DT and their possible configurations make it difficult to investigate, define, specify, and verify a consistent set of requirements related to DTs. Thus we need to extract valuable and relevant information from descriptions of DT characteristics and functionality, providing a common basis for further analysis.

5.1. What are the requirements of a DT for production systems in the O&GI usually specified or (even informally) described during DT development?

In the descriptions gathered from selected works, DT requirement definitions appear in many formats and styles, from explicit definitions to literature reviews or even informal descriptions extracted

Table 3
Categories of requirements described in the reviewed Articles.

Reference	Physical Entity	Virtual Entity	Data	Connection	Services
[33]	✓	✓	✗	✓	✓
[32]	✗	✓	✗	✓	✓
[31]	✓	✓	✓	✓	✓
[36]	✗	✓	✗	✗	✓
[53]	✓	✗	✗	✗	✓
[54]	✗	✓	✗	✗	✓
[55]	✓	✓	✗	✗	✗
[56]	✗	✓	✗	✗	✗
[57]	✗	✓	✗	✓	✗
[58]	✗	✓	✓	✓	✗
[59]	✗	✓	✗	✗	✓
[60]	✗	✓	✗	✓	✓
[61]	✗	✓	✗	✗	✓
[62]	✓	✓	✗	✓	✗
[30]	✓	✓	✓	✓	✓
[63]	✓	✓	✗	✗	✓
[64]	✗	✗	✗	✓	✓
[65]	✓	✓	✗	✓	✗
[66]	✓	✓	✓	✓	✓
[67]	✗	✗	✓	✓	✓
[68]	✓	✓	✓	✓	✗
[69]	✓	✓	✓	✓	✗
[70]	✗	✓	✗	✓	✗
[34]	✓	✓	✗	✓	✗
[71]	✗	✓	✗	✗	✓
[72]	✓	✓	✗	✗	✗
[73]	✗	✓	✓	✗	✗
[74]	✓	✗	✓	✓	✗

from known issues in the O&G Upstream industry. Such requirements provide us with what is expected of a Digital Twin in different areas of the O&GI; what follows is a description of these expectations, as well as compiled listings of such information.

In literature, DTs have been employed for various purposes, including optimization, security, monitoring, prediction, training, or enhancement of physical processes. Typically, a DT can integrate one or more of these functionalities.

Compared with typical DT of products found in manufacturing works, the O&G's DT can cover a wider range of fields, more complex models, more diverse application scenarios, and larger data dimensions and scales [70].

In the next sections, we describe the list of DT requirements identified in selected studies, i.e., it reflects what we have collectively found in the 28 studies. To explain each requirement, we establish a simple definition. The frequency of occurrences and the studies reporting each problem and limitation can be found in Table 4.

In the following sections, we outline DT requirements and properties identified across 28 studies. Each requirement is briefly defined, and the details on occurrence frequency and study references are in Table 4.

These requirements encompass both generic DT properties and industry-specific needs, particularly in Production processes within O&GI. They serve as a foundation for developing tailored DTs in this domain, facilitating reusable specifications and architectures. These widely accepted features surely provide a basis for creating configurable DTs.

5.2. Generic properties of DTs:

Usually, these properties are recognized as non-functional requirements of DT. These properties were extracted from selected works:

1. **Scalability**: It is the capability to handle a substantial amount of data and expand as the system's complexity increases. The DT architecture should be capable of handling different data scales and complexities as the system expands [33], without compromising performance [56]. Scalability is self-evident for any large-scale DT in the O&GI; if a DT wants to replicate an entire FPSO or Offshore Platform, it will potentially require large amounts of computational power, and while third-party solutions exist in terms of hardware, the DT design itself must account for large amounts of data to give the user still a pleasant experience (such as low loading times, proper navigation and such) [33,56].

2. **Modular Architecture**: It is increasingly popular within DT frameworks [36] and is fundamental for the design and use of systems composed of separate elements. These can be linked up to each other, replaced or added. Such a modular approach emphasizes separating the functionality of a DT into independent, interchangeable modules, with each module encompassing all the components required to execute a single aspect of the intended functionality. A consequence of such property is the viewpoint of a DT as a System of Systems (SoS): Such modular architecture allows a DT to incorporate legacy software. Indeed, one of the significant aspects seen in published work is a frequent need, in all domains, not just in the O&G domain, to integrate existing systems into the DT paradigm [32]. It is necessary to consider the extensive previous work currently in operation mainly due to its reliability over years of usage. Thus, companies are reluctant to rebuild such software and much effort is made to DT incorporate existing systems/programs into its operation [53]; a DT modular architectural standard would also permit a modular development approach, minimizing costs, allowing for dedicated groups to develop each module and improve efficiency than with singular DT software. [69]. It is very sound to design a DT as a modular system, as pointed out by Wanasinghe et al. [36], Thoresen et al. [69], since the DT is expected to integrate many existing solutions as well as being an ever-evolving technology that will integrate new solutions into it as both the digital and physical medium evolve over time.

3. **Interoperability**: It refers to the ability of different components of DT to collaborate and share data seamlessly and efficiently. In fact, such easiness of integration implies compatibility with various software platforms, data formats, and standards. A robust architecture supports interoperability by adhering to O&GI standards and protocols and allowing easy integration with different existing systems and technologies. For the O&GI, this is a highly desirable feature within current industrial contexts [60,61] and goes hand in hand with the DT's modular architecture efforts, especially when it comes to Data Federation solutions [53]. Higher degrees of interoperability can further enforce both technical and even semantic standardization between software and hardware developers, avoiding human and non-human miscommunication and information wastage [36]. It can connect users from different fields and areas of expertise, such as engineering simulations, control and safety systems, and training of personnel, all

Table 4
Summary of requirements identified in selected studies.

Requirement	Frequency	Group/Domain	References to studies proposing the requirement
Accurate Asset Simulation	14	Functional/ Technical	[30–32,53,54,56,58,65,67,69,71,72,74]
Real-Time Manipulation	14	Functional/ Technical	[30,32–34,36,55,58,65,69,71–74]
Big Data Analytics	13	Functional/ Technical	[30,31,36,58–61,63,64,66,71,72,76]
Integration/Standardization Between Platforms	13	Non-Functional/ Technical	[30,31,36,53,54,58,60,61,64,67–70]
Improved synergy and intra/inter-team collaboration	12	Non-Functional/ Business	[30–32,36,53,56,58–60,71,72,74]
3D/2D Modeling	11	Functional/ Technical	[31,33,34,53–55,64,65,69,71,72]
Predictive/AI Assisted Asset Management	8	Functional/ Business	[30,31,33,34,53,57,60,64,66,71,72]
Data Federation	7	Functional/ Technical	[31,33,53–55,58,64,65,72,73]
Collaborative Platform	6	Non-Functional/ Technical	[33,36,56,57,60,62]
Scenario and Operational Risk Evaluation	6	Functional/ Business	[56,58,59,64,70,71]
Visualization of Contextualized Information	5	Functional/ Operational	[53,58,61,62,73]
Process Workflow Optimization	5	Functional/ Business	[63,66–68,71]
HSE-Related Requirements	5	Non-Functional/ Operational	[56,58,59,64,71]
Training Personnel	3	Non-Functional/ Business	[60,62,69]
Prototyping of Assets	3	Functional/ Operational	[58,60,62]
Cloud and Edge Computing Capabilities	1	Non-Functional/ Technical	[63]
Domain-Specific Modeling Language	1	Functional/ Technical	[67]
Stress Testing	1	Functional/ Operational	[58]
Automatic Self Repair and Diagnostic	1	Functional/ Operational	[72]

under the same vocabulary, data structures and general standards. This fundamental requirement also benefits outsourced employees, who would also benefit from this interoperability by reducing planning costs and unexpected complexity from unknown or unplanned systems [31,54,64]. To achieve higher degrees of Interoperability [58] recommends an equally integrated and varied team of personnel involved in the development, emphasizing the importance of building upon existing consolidated non-DT systems, particularly to avoid wasting legacy work in achieving interoperability, when introducing the DT paradigm is crucial [69].

4. **Easiness of use** is a basic concept that describes the ease with which users can interact with a product, and it is sometimes considered synonymous with usability, related to the ease of use of digital technology, and is frequently employed in UX design processes. In the context of a DT, easiness of use is achieved by an intuitive and user-friendly interface for interacting with the DT. Visualization tools are a way to provide such easiness of use, making insights accessible to various stakeholders. Moreover, advanced visualization tools make it possible to present complex data and simulations in an easily understandable manner [58]. Ease of use is especially important in the DT context due to the wide range of operational personnel with different backgrounds. In an ideal DT, it might be necessary to create different interfaces or visual paradigms to access the same DT so that they are tailored to different types of users, from a managerial human resources employee to a maintenance engineer [62,73].

5. **Cybersecurity**: It is an essential characteristic of a DT, defining robust measures for protecting connection networks, devices, assets, and data from unauthorized access or criminal, and consequently ensuring confidentiality, integrity, and availability of information. It is critical that end-to-end security solutions are integrated to help clients secure, test, and monitor operating environments and raise organizational maturity as a prerequisite for digitalization [31]; due to its Real-Time and (usually) internet-enabled operations, extra care must be taken when applying cybersecurity features to a DT [36]. The DT should make extensive use of current state-of-the-art cybersecurity technologies and user access mechanics to provide sufficient security to the user (white & black listing, restricted access to different user groups, etc.) [62,63]. One significant hazard linked to DT is that they can provide cyber-criminals with a new way to access and exploit sensitive information, which could be used to plan and execute a targeted cyber attack. Cybercriminals might employ DTs as a means to infiltrate and target the real-world physical assets they replicate. In the case of a DT representing a power plant, unauthorized access could be gained to manipulate the plant's controls, posing a risk of damage or disruption. In extreme cases, this could disturb crucial services and threaten human, animal, or environmental well-being. O&GI (or any

organization with sensitive information) takes several approaches to secure its data and assets, and thus, the DT platform should support and integrate itself into existing cyber-security approaches [33].

6. **Human Risk Reduction**: The cybersecurity strategy remains incomplete without addressing the human element sufficiently. We must equip employees with the tools, knowledge, and skills necessary to make the right decisions and minimize opportunities for breaches, thus mitigating risks associated with human factors [56]. One of the principal proposed reductions in human risk is the capability of much high-fidelity training: in addition to traditional training, a DT could even simulate high-risk situations. The users can train in these risky situations and better prepare themselves in otherwise very difficult situations to train on realistically [58,59]. There is also a significant study on DTs improving situation awareness within the work environment [73], providing workers with a consolidated visualization of workflow processes, thus allowing quick evaluation of their current tasks.

Another effective risk reduction method involves employing predictive AI techniques, for example, in order to proactively prevent and address issues as they arise [64,71].

7. **Cost Reduction**: In a company, it involves determining and eliminating expenses that do not contribute value to customers, while optimizing processes to enhance efficiency. Typically, cost reduction is centered on achieving immediate cost savings. Many of the other requirements also contribute to cost reduction, such as preventing cyber-attacks with an effective cybersecurity scheme [36]. Cost Reduction also follows the same pattern as Human Risk Reduction, as some of the same benefits attained by DT-assisted personnel training and predictive capabilities can reduce costs as well [56] by reducing life-threatening situations and making the training of personnel a more streamlined pipeline [58,60]. Global green initiatives and HSE concerns in the production process seemed threatening to the O&GI — but O&G companies are investigating how to cut costs and stay competitive. Some works propose that changes to work culture might be needed, such as changing the criticality of CAPEX levels when considering DT projects [36,56]. Other works find that advanced technology can be the key to cost reduction [60]. In fact, O&G enterprises are adapting their operational strategies to embrace technological advancements, leading to a reduction in overall costs. Separated adoption of smart machinery, incorporating artificial intelligence (AI) advancements into traditional equipment and cloud-based solutions, for instance, may help boost their production and their bottom line [58]. However, implementing solutions that integrate all these technologies – such as a digital twin – tends to have an even better result. For example, articles such as [32] argue that managers tend not to evaluate projects for long-term effects and that the cost-saving effects of a DT implementation should receive much higher priority.

5.3. DT requirements obtained from selected works

The surveyed literature revealed different requirements related to DTs. This section summarizes the properties and requirements for DTs as described in the selected studies. In Table 4, this information is presented in a more compact format: for each requirement, we identify its category, the frequency of occurrences and also the studies reporting the requirement. Furthermore, we also have categorized these requirements into groups, concerning their scope, the three groups being:

- Business: Requirements related to the overall business goals, providing innovation and/or strategic advantage with the use of the DT.
- Operational: Requirements related to the day-to-day operation and maintenance of the digital twin.
- Technical: Requirements focused on the technical aspects of the system, including infrastructure, software, and hardware.

While some references may use terms like “strategic”, “tactical”, or “system-level” instead of “business/operational/technical”, the underlying principles align with this categorization [77–79]. For example:

- **Business** is similar to Strategic/economic goals.
- **Operational** is similar to Process/workflow efficiency.
- **Technical** is similar to Infrastructure/data enablement.

Thus, here we describe the list of 19 requirements identified in selected 28 studies.

The most frequently mentioned requirement is **Accurate Asset Simulation**, which reflects one of the core ideas behind the Digital Twin concept [32]. 14 out of the 28 studies strongly focused on discussing and emphasizing that this requirement is of utmost importance. This may seem the obvious goal of a DT, but there are many obstacles, including even the language itself being used in the DT design [80], which force necessary adaptations and shortcuts when translating a physical entity into its digital equivalent, right down to even its knowledge graphs or documentation. In fact, if confusion and loss of fidelity can already happen at such a fundamental stage of modeling a DT, a worse loss can be expected at later stages of development [64]. This predominant requirement cements the need for a Digital Twin that represents, at a highly acceptable level, the desired physical entity. The solutions presented are varied: from the usage of an organized and developed study of the physical entity, with supporting knowledge/language graphs, to the development of domain-oriented ontologies [81] or customized methodologies for development [58,60]. Clearly, all solutions share the goal of reducing the potential loss of DT fidelity.

In addition to accuracy in mirroring the physical world, the ability to simulate assets and processes is also relevant. Simulation is also of concern for a diverse number of industries (including O&G) [60,62]; these simulations vary in granularity and know-how required; they can be actual physical simulations (requiring very in-depth physical understanding) [53] or just simulations of processes (made by either machines or humans) [33]. Simulation is very important for many processes of O&GI and can be used in many situations within the DT framework. Konchenko et al. [54] discusses usages of 2D, 3D, or even virtual reality to create a synchronized well site representation whereas Thoresen et al. [69] presents modeling and simulation of all sensors, actuators, valves, and drives which the Managed Pressure Drilling control system is connected to, making easy to investigate faults in different components. A cloud-based simulation platform (with Python-based scripting) is the basis of a framework to automate the continuous tracking of the assets’ structural integrity and assessment of their current state [72]. Even though simulators can run without any feedback from sensory data, further data could be extrapolated using machine learning techniques [59], reducing overall risks, from maintenance to general equipment failure.

Real Time Data Manipulation is the second most mentioned requirement - with 14 out of 28 papers. It is an industrial requirement and encompasses the spectrum of real-time remote control and monitoring, software-assisted monitoring [30,33,55,65,71–73], data analytics [30,53,55,58,65] as well as some simulation capabilities [32,65,67,69,71,74]. Indeed, real-time access to quantitative data and advanced analytics are both already present in many existing monitoring systems facilitating efficient and more informed decision-making [72]. So it is a naturally important requirement for a DT. A production process is usually monitored by sensors that collect, store, and process plenty of data in real-time, aiming to use such data to anticipate events [33,73]. Although it is quite difficult to monitor a very large system in real-time physically, a DT can be monitored and remotely controlled, reducing the need for physical presence in hazardous and unhealthy environments [36]. The O&GI usually adopts some system for gathering, enrichment, storage, and access to reliable, real-time operational data, either for predictive operation and maintenance or for enabling a 360°

real-time view of operations across the value chain. Indeed, readily available real-time information coupled with access to relevant historical and recent data keeps stakeholders informed, thereby enhancing transparency in the decision-making process [33,36,56,59,72,73].

A highly mentioned requirement – closely related to interoperability property, seen in Section 5.2 – is **Integration/Standardization Between Platforms**, the integration of already existing automated systems into a DT Framework. The reasoning is quite evident: the reuse of the (usually) huge knowledge put into legacy modeling/simulation software and hence the saving of related time/effort [31,36,67]. For example, some existing systems are management systems for specific equipment (such as offshore platform valve pressure control, flow control, and FPSO modeling) or even personnel management software. Consolidating and unifying these systems into a single platform was already a previous O&GI desire; the DT platform provides even more attractive motivation and an environment to implement these ideas [53,54].

Another very cited requirement is the need for **3D/2D Modeling** capabilities – also named as **Digital representation of assets** by some studies – mentioned by 11 out of 28 papers. Since many modeling software (such as CADs, BIMs, etc.) [34,64] is already integrated into systems within many of the industries interested in DT technology, it is related to the "Integration/Standardization Between Platforms" requirement. However, modeling integrated into most DT frameworks and paradigms is more complex than modeling a standalone software offers. [31,53,54]. Such integration requires active coordination with sensors composing its infrastructure and correct throughput of data for real-time needs [33]. Ideally, it also requires communication between the DT layer where modeling is being done with its actual physical counterparts in order to play a crucial role in optimizing operations, safety improvement, and enhancing decision-making in the O&GI. Basic representations of DTs, usually generated during the project phase, are populated by engineering data to create 1D, 2D, and 3D asset models, including the operational characteristics and behavior of the asset. If the asset is a type of equipment, such digital representation is called Equipment Model. An equipment model is a structured but flexible hierarchy of resources: a modular representation of a complete production organization in a physical and functional context. Each resource can perform a finite number of specific, minor processing activities, sometimes containing a defined set of equations with stream connections coupling these sets of equations together into a larger system. In a DT, the equipment model may be configured to recreate real-life events by using recorded data from measurements and control system inputs. An example of 3D modeling for flow assurance is presented by Mahalingam and Arsalan [55], focusing on the essential data for upstream and downstream piping, where measurements, risks, and maintenance information are the critical information.

Predictive/AI-Assisted Asset Management, cited by 8 out 28 papers, is another well-known Industrial DT requirement existing in the O&G environment as well [72]. Predictive-Based Asset Management is a reliability strategy to evaluate the operational state of equipment, aiming to identify the necessary actions for enhancing system performance and dependability [34]. AI-assisted predictive asset management assesses numerous variables indicating an asset's present condition, forecasts outcomes based on usage patterns, and alerts maintenance teams to potential equipment failures beforehand [30,53]. AI-assisted asset management can significantly improve industrial performance in current operations and future optimization through predictive capabilities [33].

In order to provide the Asset Information Management, a centralized repository of asset information enables users to capture, connect, access, and visualize enterprise-wide data from several possible sources and formats, such as documents, drawings, registers, 3D models, and databases [33,36,53]. It also gives users control of their project and asset information throughout the entire asset life cycle [59]. The Asset Management functionalities of a Digital Twin entail the diagnosis, monitoring, and maintenance of assets, encompassing electrical supply/distribution, process equipment, and the automation infrastructure responsible for monitoring and controlling these assets. In O&GI, this allows workers to answer questions such as: "How critical/dangerous is a task?", "What other tasks depend on it?", "What are the other people involved?", which implies better safety in the work environment. Asset performance management involves gathering data on equipment performance and offering tools and applications to predict the optimal utilization of assets, ensuring the attainment of operational goals while minimizing costs [53].

As the digital twin's life cycle progresses, it must be updated in real-time with operational data such as risk matrices, spare parts inventory, maintenance records, and known physical behavior across operational functions [31]. Feeding the digital twin historical and real-time data requires a data historian that collects, stores, displays, analyzes, and reports on operational and asset health information [64]. A data historian also referred to simply as a historian, is a specialized database engineered to gather and archive time-series data from diverse sources within a process plant [53]. It autonomously records process and production data, compressing it for more streamlined storage. A historian can make data accessible to a broader audience in the organization, enabling them to conduct equipment performance analysis, streamline operational processes, and improve asset health. Along with a historian, an enterprise asset management solution is critical to provide centralized maintenance, procurement, and inventory management for assets [53,65].

Predictive and scheduled maintenance is very related to asset management to prevent failure in the future. Since multiple sensors monitoring physical assets generate plenty of data in real-time [31,33,57], system failures can be detected well in advance through intelligent data analysis, which will lead to better maintenance scheduling, reduced risks to human life [53] and reduced downtimes of operations. In the O&GI, operators are realizing that DTs, if harnessed effectively, can help dramatically shrink their CAPEX and OPEX [60].

Data Federation, cited by 7 out of 28 papers, focuses on centralizing all heterogeneous data from different sources already present in the company's existing tools and software, a diverse pool of data types and patterns, in a single, centralized repository [7]. As a fundamental part of a DT's integration effort, this requirement emphasizes trying to standardize data beyond a single software or supplier, thus avoiding specific solutions becoming standards. More, a DT should provide **processing and storage of data in different formats and from distinct sources** [53,54]: real-time data, series of historical data, data created by simulation, data obtained by physical modeling, data generated by some AI or Machine Learning tool, etc. This also extends to data from radically different equipment and in different scales, from platform motors and flow measurers [55] to entire naval entities such as an FPSO [33]. Such data are used as input parameters of several services working with different data formats that cannot simply be reformatted for historical or practical reasons, like data correlation and time series alignment analysis, visualization, and model optimization.

Big Data Analytics refers to an intricate process of analyzing extensive and varied data sets comprising structured, semi-structured, and unstructured data from diverse sources and of varying sizes in order to reveal valuable information, including insights, obscure patterns, correlations, market trends, and customer preferences. In the DT for O&G domain, this enables complex applications like proactive condition-based maintenance (CBM) [60], intelligent monitoring [30], what-if analysis, or predictive maintenance [53], or even to uncover anomalous asset behaviors or some changing environmental conditions [61].

The need for a **Collaborative Platform**, mentioned in 6 out of the 28 analyzed studies, refers to the DT having the capability of integrating human activity into the same platform as other asset management [33]. The DT should also serve as a communication platform to share work and to report progress on assignments, as mentioned by McLachlan [56], enhancing synergistic interactions and collaborations. With increased autonomy and access to all pertinent information, teams achieve **improved synergy and intra- and inter-team collaborations** – another mentioned DT requirement – improving productivity [36,59].

Undoubtedly, a DT should facilitate the analysis of hypothetical (what-if) scenarios, leading to an enhanced risk assessment [56,58,59]. This requirement – **Scenario and risk evaluation** – enables the synthesis of unforeseen scenarios, examination of the system's responses, and corresponding mitigation strategies, and it is also considered essential by 6 studies. Countless operational risks confront operators in the O&GI, where a swift comprehension of appropriate actions in the event of an issue at the industrial plant can determine whether an accident occurs or not [70]. Such analysis provides personnel with better training for potential scenarios, all without compromising the integrity of the real DT's asset [64,71].

A particular set of DT requirements in the O&GI is formed by **HSE-related requirements**, mentioned in 5 out of 28 studies. In every domain, the usage of a DT should reduce risks regarding safety and security in dangerous work environments [56]. In fact, the O&GI is one of the most complex sectors to protect [64]: the nature of upstream oil extraction lends itself to potentially very costly ecological disasters, and these incur not just monetary costs due to governmental fines, but also potentially catastrophic health costs to people and wildlife. With this in mind, it is no wonder O&G companies have a vested interest in using DTs to minimize these risks. A multitude of distinct risks looms (e.g., hazardous working environments exposing employees to dangers, remote and vast locations, poor perimeter security, internal sabotage, etc.), posing threats to both safety and operational stability while putting essential resources at risk. Ensuring comprehensive physical security coverage can enhance the safeguarding of internal and external spaces, fortify vulnerable zones, and aid in compliance with regulations and legislation. Various physical security technologies are available for deployment in O&G pipelines and related above-ground infrastructure to elevate their security measures and tackle all challenges [58,59,71]. However, it is crucial to integrate both physical and cyber security for optimal outcomes. Safety is also an important concern, from training to daily activities. As mentioned in Douglas and Rios [32] and conceptualized by Umeda et al. [82], a DT can be used to train workers through a digital triplet concept: essentially a specialized twin of a work environment oriented towards teaching/learning for employees. Through this environment, situations can be simulated with higher accuracy of all resources involved, using the same tools as their real digital twin counterparts.

Some HSE requirements that were briefly summarized in Section 2 are:

- Assess HSE considerations before initiating new activities and conduct re-evaluations when significant changes occur;
- Require that contractors working on their behalf strictly adhere to HSE standards that completely align with their own;
- Acknowledge the concerns of employees and society regarding HSE matters. Offer them continuous training and pertinent information, actively engaging them in discussions concerning company policies and practices in these domains.
- Define and uphold contingency plans, jointly with emergency services and authorities, to mitigate damages in the event of accidents.
- Establish collaboration with governments and other stakeholders to enhance regulations and O&GI standards concerning HSE matters.
- Conduct or support research endeavors focused on enhancing the HSE aspects of their products, processes, and operations.

An essential component of a DT provides **Visualization of contextualized information** [58], delivering universal access to desired information wherever teams are located. The desired information may be a 360-degree view of an enterprise asset management tool, of data in the asset information repository, of historical data or even data resulting from predictive analytics. Visualization allows a perspective of all the components together and, with appropriate filters, promotes rapid, collaborative, and effective decision-making [61,62,73]. Although not focused on upstream, a description of DT visualization features successfully resulting in substantial cuts to maintenance costs in a real-world O&G company is presented by the study [53].

A DT can make use of all its data as the basis for **Optimization**, employing advanced techniques to analyze them and then to enhance system performance, efficiency, cost-effectiveness, risk mitigation, and decision-making capabilities. [68]. For this, a DT should establish clear, measurable system objectives along with corresponding cost functions to regulate and assess their achievement [66]. The study [63] describes upstream applications where a DT can intervene, such as analyzing production rate, equipment performance, and failure rates, whereas [71] shows DTs identifying and optimizing possible bottlenecks alongside preemptive solutions to maintenance problems.

Due to the simulation nature of the digital asset, a DT can be easily used as a platform for **Training Personnel** [60,62]. Umeda et al. [82] recognize the importance of a DT platform for training new employees and retraining old ones. The DT can simulate a work environment: it can be used by trainees to familiarize themselves with normal operations, but mainly with abnormal operations. In fact, the latter are much more difficult to train on within a non-digital assisted training environment.

Two simulation-related requirements are **Prototyping of Assets** and **Stress Testing**. With DT prototyping, much of the costs and wasted effort can be avoided by correctly simulating changes such as new equipment installation, different asset parameters or even new configurations of work environment [60,62]. Prototyping of assets is also useful as a support for construction. A DT should not only be a tool to help create and design the physical assets but also offer support in evaluating such actions; by considering available information, it could offer evaluations on proposed efficiency gains or investment needed for such new equipment/construction [71]. Following the changes, it is advisable to do stress testing [58]. With the aforementioned predictive capabilities of the DT, stress tests can simulate close-to-real stress situations on various assets.

Cloud and Edge Computing is an infrastructure requirement [63]: the DT should be ready to outsource the processing of results to its edge nodes, not only as a performance issue but as part of the already-mentioned modular architecture. Although [21] claims cloud and edge adoption in the O&GI was late compared to other industries, mainly because of security and data privacy concerns, cloud, and edge computing provide greater storage and computational capabilities, bypassing the need for a robust local IT infrastructure.

A **Domain-Specific Modeling Language** for developing applications in O&GI [61] – representing physical components, non-physical components, and the interactions between them – is fundamental to represent this knowledge and semantically integrate the functions of the DT [80]. In many cases, such semantic integration enables interoperability between the DT services and also the integration between people, reorganizing the common vocabulary used in O&GI. [80] states that if users and DT can share the same language and concepts, their use will be more integrated and accepted.

A last requirement from selected studies is **Automatic Self Repair and Diagnostic**. According to Podskarbi and Knezevic [72], if a DT can achieve the goal of automatically self-repair, it is fully integrated with its physical counterpart, and thus DT concept as a whole would quickly spread throughout industries.

All the requirements described in this section were identified and discussed in selected studies. According to the literature review, the requirements of a DT for a Production System in the O&GI include several of these characteristics, from generic properties like scalability and easiness of use to complex and sophisticated procedures like predictive maintenance, real-time remote monitoring, and adoption of visualization tools. Since they correspond to core DT functionalities, a DT prototype or proof of concept could be developed from such requirements.

6. Discussion and conclusions

We investigated the requirements of DTs, with a particular emphasis on the production systems of the O&GI. We reviewed 28 studies that addressed this topic. Our research protocol defined the research question aimed at systematically exploring the explicitly stated requirements for DTs. Finally, as a possible answer to the presented RQ1 – "What are the requirements of a DT for Production Systems in the O&GI usually specified or (even informally) described during DT development ?" – was the list of essential Requirements of Digital Twins for Production Systems in the O&GI as presented in Table 4 and explained in Section 5.3. Such a list may be used as a starting point for requirements of Digital Twins for Production Systems in the O&GI during the implementation of new Digital Twins or as evaluation criteria for existing Digital Twins.

In this section, we present threats to our SLR's validity, summarize our work's main contributions, and present some possibilities for future work.

6.1. Threats to validity

A threat to validity is described as a self-evaluation of potential problems that might compromise the research [83]. Firstly, we highlight a risk related to our search process since the ability to find relevant studies directly impacts the results of the SLR. To prevent such risk, we use the snowballing technique, as proposed by Jalali and Wohlin [84]. Nevertheless, we may have some relevant, unrecognized studies in our search process.

Another risk is the selection process. Researchers apply inclusion, exclusion, and quality criteria to obtain relevant studies without in-depth evaluation, which can lead to false positives and negatives. To mitigate this, at least two researchers reviewed each article.

We also observed a risk in data extraction. Researchers perform qualitative analyses to answer their research questions, and their conceptual background and interpretation skills can affect decisions and compromise results since some articles on DT requirements also lack detailed definitions. Understanding the motivation or source of these requirements is crucial since they may vary based on context, DT configuration, or objectives. These variations can lead to inaccuracies in the data extraction process. It is categorized as follows:

- Internal Validity: To avoid compromising the study from internal factors and bad decisions, we use a consistent and proven methodology (see 4), rigorously applying the exclusion and inclusion criteria in all articles and evaluating them with an equally rigorous and methodical search for the RQ presented.
- External Validity: Concerns the validity of the generalizations made in this paper. Similar to internal validity, a systematic process (see 4 and alongside 4 and Figs. 5, 3) was used to retrieve the information presented, which identified natural biases that arise from the chosen publications.
- Conclusion Validity: It analyses how interlaced the treatments and the outcome of SLR. To this, we used the Quality Assessment standard described in 4 and shown in Table 1 and 2.
- Construct Validity: This item relates to the theories, concepts, methods, and measures the researchers have in mind while conducting the research and interaction between them and possibly other simultaneous projects they work on. We established an evaluation method in 4 while trying to keep our study's validity in mind, including the impartiality demanded by this type of validity.

6.2. Main contributions

The concept of DT can potentially improve performance and productivity in the O&GI through effective decision-making enabled by real-time monitoring, predictions, simulations, and optimization of processes. This study conducted a systematic literature review to identify the requirements of a DT for the O&GI, especially for the production system. Consequently, the study identified some research gaps and ideas for future work in this field. The paper's main contribution is identifying requirements described in existing research literature. Such a collection of state-of-the-art requirements provides guidance and feedback for developing DTs for the O&GI. In principle, no technology provider can meet the wide range of requirements related to a digital twin. However, even if a single supplier met all requirements, several experts say relying on a single technology provider is not advisable. Since digital twins could be created by integrating technologies and tools that were developed independently, accurately capturing and representing these requirements is a pivotal task for such integration.

The requirements engineering process for building novel DTs or modifying existing DTs requires future research on techniques and processes. The research literature lacks mature, practical applications and a more complete description of the industry.

According to LaGrange [30] all O&GI has been slow to adopt digitalization and other industry 4.0 solutions, mainly due to the lack of a solid foundation for solutions. Certainly, a better definition of the requirements of such solutions, specifically DTs, is an important contribution to this foundation.

This work addresses requirements independently of technology and implementation: a requirement is an abstract description of "what" a DT should behave like. However, successfully implementing digital twins in practice requires knowledge about technology (networks, programming languages, AI models, etc.) to make these requirements real by addressing "how" a DT can be built to meet specified requirements. Ideally, a DT design and implementation should be preceded by a requirements specification, and the DT requirements team and the DT construction team have distinct knowledge and viewpoints.

This survey was developed in 2023, and the results reported cover studies between 2016 and 2022. Thus, many relevant, more recent studies published since then are not included. Nonetheless, we are convinced that our contribution is still pertinent for those who are working on DTs in the O&GI.

6.3. Future work

According to our SLR, the DT community has been increasingly studying topics related to understanding DT requirements. However, recognizing and understanding the requirements described in the studies on DTs was a challenge as not all requirements, properties and characteristics have the same relevance in all types of DTs, particularly in the O&G domain. More, very few papers describe the RE process in detail, perhaps due to the confidentiality of companies [68]. Thus an effort is required to document these requirements in a more standardized and agnostic manner.

Maintaining cutting-edge knowledge about DTs is a continuous effort because the number of studies is increasing significantly. We recognize that the analysis of new studies as future work is very important, particularly to improve understanding of the DT requirements.

Driven by a growing interest in the field, a significant lack of literature dedicated to DT requirements is noteworthy. This gap is apparent when examining the methods employed for eliciting these requirements, the notations utilized for their representation, and the scarcity of tangible, real-world examples illustrating DT requirements. Our SRL identified the need for more research on DT requirements and the RE process for DTs, including elicitation techniques, representation notations, and validation techniques.

Our work reports that very few, among all selected studies, explain or report on the techniques used for requirements gathering and representation. Indeed, many of the works presented here seem to use just literature definitions as a starting point for their DT requirements and, despite the complexity and impressive results in their fields, do not document what happened during the RE process. Remarkable exceptions are Lau et al. [71], Bo et al. [68], presenting where and how their requirements were obtained: traditional interviews with experts in the O&G field are their primary source of requirements gathering.

Although DT requirements specifications traditionally describe functionality, our RSL has found some descriptions of quality properties, use scenarios, and user interfaces of intended solutions. Further study is also necessary to identify if notations commonly used for requirements representation, such as use cases, user stories, agents, and goal-oriented languages, could be considered useful for the specification of DT requirements. Sometimes, a combination of notations could be interesting, for example, natural language can be complemented with UML diagrams. In this RSL, natural language predominated as the adopted notation.

Adopting some technique for validating DT requirements is just as critical as adopting techniques for gathering them. Ensuring that the identified requirements are accurate, feasible, and aligned with stakeholders' expectations is essential to the success of a DT.

Thus, in addition to our systematic review focused on requirements understanding, it would be interesting to identify the gathering techniques adopted for eliciting/discovering such requirements, what the requirements sources (users and other stakeholders, documents, procedure manuals, enhancement requests for some existing system, and even marketing material and product definitions), and also the way requirements are represented (from informal expression to more rigorous models). However, existing works in the DT literature do not discuss the adoption of requirements-gathering techniques or notations for requirements expression.

Although the focus of our SLR was not to analyze tools to support the RE process for DTs in O&GI, this subject should be better investigated. This would allow researchers to understand how such a process is planned and supported and what can be done to improve it.

The continued advancement of DT relies on developing industry-specific standards that align with the unique characteristics of each sector. Knowledge of and compliance with applicable standards, regulations, and laws [85] helps a company in (1) maintaining the safety of its facility, (2) adhering to process safety practices consistently, and (3) mitigating legal liabilities. Ensuring operational integrity necessitates compliance with state, federal, and international regulatory standards. This involves adhering to various standards and regulations, including [86], audit trails, KPIs, etc. Regardless of the company's size, adherence to these laws is imperative for fostering accountability and sustaining a competitive edge.

Despite the insights about DT requirements achieved in this work, mainly concerning requirements related to the O&GI, the road to a consensual definition of requirements for such an industrial context is still long. First, we must validate such requirements with domain experts in several distinct contexts. Since not all requirements are equally important, it is important to identify and manage the priority of various requirements, depending on the limited resources available and the objectives of each DT. Finally, we need the development of Proofs of Concepts (PoCs) or actual DTs built totally or partially based on this requirement set. Since most of the current works are theoretical or present partial implementations, we would like to establish a common, well-defined and unambiguous set of requirements serving as a basis for applications constructed by distinct suppliers or operators in this domain. We also aim to make requirements evolve by adding new features and modifying existing features, a natural phenomenon in the development life cycle of requirements in any system.

In the near future, we will witness the emergence not only of groundbreaking new technologies related to digital twins but also the ubiquitous adoption and integration of Digital Twins in many sectors of the O&GI.

CRediT authorship contribution statement

Ricardo C. Belo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Marcelo S. Pimenta:** Supervision, Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Tarciso T. Salvador:** Data curation. **Rafael H. Petry:** Writing – review & editing, Visualization. **Mara Abel:** Project administration, Funding acquisition.

Declaration of Generative AI in Scientific Writing

The authors declare that they have used ChatGPT (GPT-3.5) exclusively as a tool to improve existing readability and language; this was done with human oversight and control, as well as proper reviews and edits of its results.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ricardo C. Belo reports financial support was provided by Libra Consortium. Marcelo S. Pimenta reports financial support was provided by Libra Consortium. Rafael H. Petry reports financial support was provided by Libra Consortium. Mara Abel reports financial support was provided by Libra Consortium. Ricardo C. Belo reports financial support was provided by FINEP — Financiadora de Estudos e Projetos. Marcelo S. Pimenta reports financial support was provided by FINEP — Financiadora de Estudos e Projetos. Rafael H. Petry reports financial support was provided by FINEP — Financiadora de Estudos e Projetos. Mara Abel reports financial support was provided by FINEP — Financiadora de Estudos e Projetos. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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